

Folia: Final Presentation

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Problem and Vision

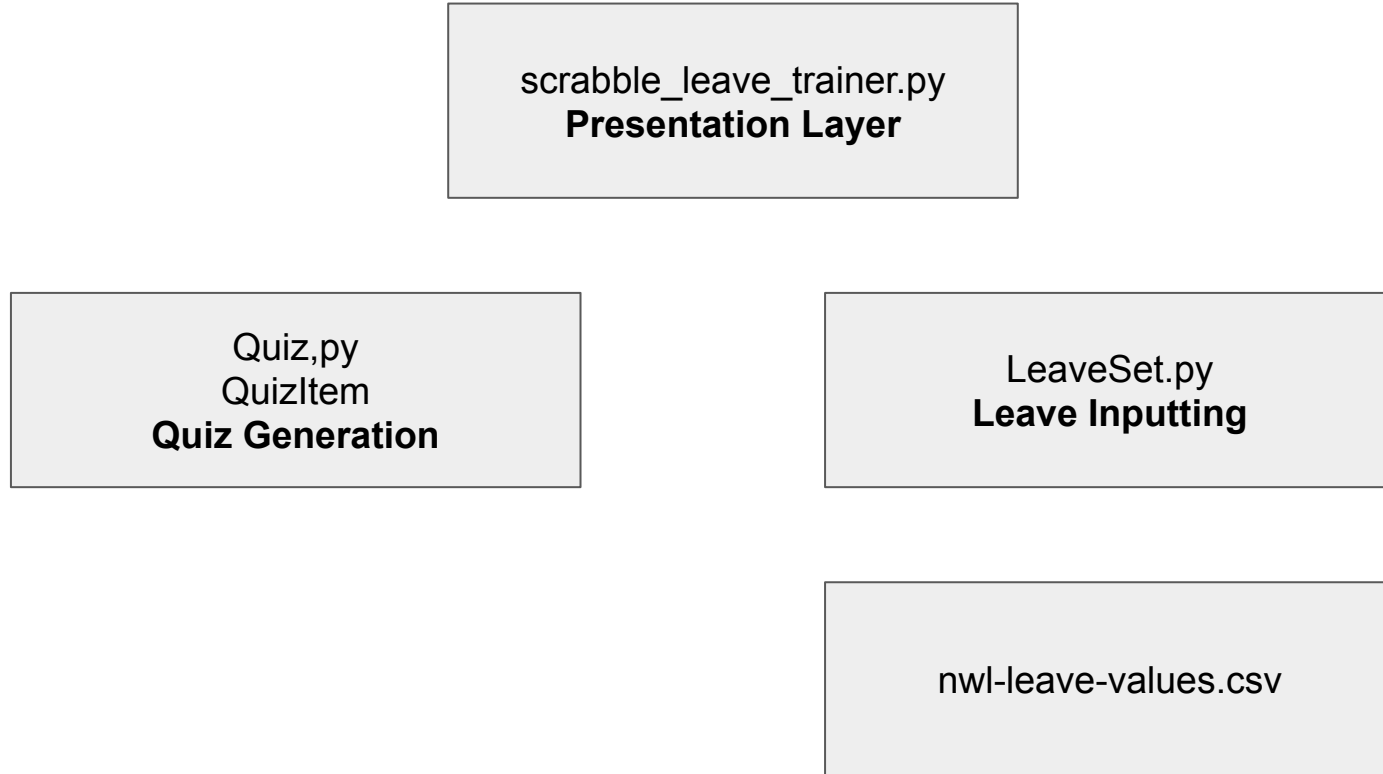
- Scrabble players need to evaluate the “leaves” that they face (tiles remaining after a play)
- There are 900,000+ leave combinations that have strategic values in a range
- You cannot memorize all these values, so it is best to develop intuition
- We propose to make a **desktop study tool combining interactive quizzes with instant lookup** to develop this intuition



What Distinguishes Folia

- This is the first leave based application relating to this Scrabble study method
- This uses active learning in terms of quiz-based training rather than solely using a lookup feature or passive reference
- There are customizable filters and difficulty levels based on the quiz settings
- There are zero dependencies, so it is pure Python and Tkinter

Architecture



Lessons Learned

- Performance optimization was difficult, and it took time to load the 900k CSV
 - This required optimized parsing and pre-filtering, which reduced the time
- GUI responsiveness took a while to fix, as the GUI kept freezing
 - From this, we separated data operation from a UI, so our data could load
- It was also important to validate the inputs that we had, both from the CSV and from the users
- In general, it was best to have simplicity to negate feature creep, so focusing on polishing functionality before adding any other features and not over-adding features